

ILAK MANOHARAN

(Legal name: Ilakkuvaselvi Manoharan)

Full Stack Software Engineer

Visa Sponsorship Not Required

LinkedIn | GitHub | Medium | YouTube

PROFESSIONAL PROFILE

Experienced and results-driven Software Engineer with over 20+ years of full-stack development expertise across diverse industries including **Banking/Financial Services, E-commerce, Automotive, Heavy Equipment, Technology, Cloud Services, and AI**. Adept at architecting and delivering scalable, cloud-native applications using Java, Spring Framework, Python, Swift (iOS), and modern data engineering stacks. Proven leader in AI/ML innovation, with a deep understanding of the software development lifecycle and a track record of driving technical excellence across global teams.

- **Full-Stack Development Leadership:** Expert in delivering enterprise-grade applications using Java, Spring Boot, Python, and Swift. Skilled in both backend and frontend design, integrating scalable APIs, microservices, and modern UI frameworks.
- **AI & Data Science Expertise:** Specialized in machine learning, deep learning, and generative AI, including large language models and retrieval-augmented generation (RAG) pipelines. Experienced with TensorFlow, PyTorch, and end-to-end model lifecycle management.
- **Cloud Platform Proficiency:** Hands-on experience deploying and managing applications across AWS, Azure, and Google Cloud. Skilled in containerization (Docker, Kubernetes), CI/CD pipelines, and DevOps best practices.
- **Data Engineering and Architecture:** Strong foundation in building reliable data pipelines and scalable architecture for analytics and real-time decisioning, using tools such as Spark, Airflow, and cloud-native data services.

AREAS OF EXPERTISE

Programming Languages	Java (7/8/11/17), Python, Swift, JavaScript, TypeScript
Web & Mobile	iOS, Flutter, React Native, React.js, Next.js, Tailwind CSS, Node.js
Backend & Frameworks	Spring Boot, Spring Cloud, Spring Security, Hibernate, REST APIs, gRPC, FastAPI, Flask, Microservices Architecture
Databases & Data Eng.	SQL (PostgreSQL, Oracle, Sybase, SQL Server), NoSQL (MongoDB, DynamoDB, Cassandra), Data Warehousing, ETL Pipelines, BI Tools
Cloud & DevOps	AWS (EC2, S3, RDS, Lambda, SQS, SNS), Google Cloud (GCP), Microsoft Azure, Firebase, Docker, Kubernetes (EKS, AKS, ECS), Terraform, CloudFormation, PCF Concepts
Security & IAM	OAuth 2.0, OIDC, SAML, RBAC, JWTs, IAM, Vault, Secrets Management, Policy-as-Code
CI/CD & Tooling	GitHub Actions, Jenkins, Git, Jira, Confluence
Testing & QA	JUnit, Mockito, Selenium
Observability	Grafana, ELK Stack, AppDynamics
AI/ML & Data Science	TensorFlow, PyTorch, Pandas, Scikit-Learn, Data Modeling, Deep Learning, Machine Learning Algorithms, RAG, Generative AI Models
Methodologies	Agile, Scrum, Waterfall, Six Sigma

EDUCATION

Master of Science, Electrical Engineering — Texas A&M University – Kingsville

CERTIFICATIONS

Oracle Certified Associate: Java SE 7 Programmer • OWASP Top Ten Certification • Six Sigma Yellow Belt • Data Science in Python • Red Hat Delivery Specialist – PaaS Administration • IBM – Introduction to Data Engineering

PATENT (NOT GRANTED)

Smart Restaurant powered by Cloud-Native IoT AIOS, Pub. No: US 2024/0273653 AI | Pub. Date: Aug. 15, 2024

PROFESSIONAL EXPERIENCE

MERCOR

Remote

AI Systems Engineer / Full Stack Software Engineer

Sep 2025 – Present

Worked on AI-enabled software systems, evaluation workflows, scalable backend infrastructure, and full-stack product development supporting AI-driven platforms and operational tooling. Contributed to the development of production-grade systems integrating LLM workflows, automation pipelines, and distributed cloud infrastructure.

AI Systems, LLM Workflows & Evaluation Infrastructure

- Built and optimized AI-assisted workflows leveraging LLMs, embeddings, semantic retrieval, prompt engineering, and automated evaluation pipelines.
- Developed scalable backend services and APIs supporting AI-driven applications, candidate workflows, and operational tooling.
- Worked with structured and unstructured datasets to improve AI evaluation quality, ranking systems, and workflow automation.
- Designed systems integrating Claude, OpenAI APIs, and AI-assisted tooling into internal operational and product workflows.
- Assisted in building reliable evaluation and feedback systems for model outputs, response quality, and workflow optimization.

Backend Engineering & APIs (Python, FastAPI, Node.js, TypeScript)

- Developed backend services, REST APIs, and asynchronous workflows supporting scalable enterprise applications.
- Built data processing and orchestration pipelines for automation, integrations, and AI-driven operational systems.
- Implemented authentication, authorization, logging, observability, and secure API communication practices.
- Optimized application performance, scalability, and maintainability across distributed environments.

Cloud Infrastructure, Automation & DevOps

- Worked with AWS cloud services, Docker, Kubernetes, CI/CD pipelines, and infrastructure automation tools for scalable deployments.
- Built and maintained automation workflows integrating APIs, webhooks, and internal operational systems.
- Improved monitoring, logging, reliability, and deployment workflows using observability and DevOps best practices.

Frontend & Product Engineering (React, Next.js, TypeScript)

- Built responsive, high-performance frontend applications and internal operational dashboards using React, Next.js, and TypeScript.
- Developed reusable UI components and product workflows to improve usability, operational efficiency, and scalability.
- Collaborated across engineering and product teams to rapidly prototype and iterate on AI-enabled features and tooling.

JPMORGAN CHASE

Chicago, IL

Full Stack Software Engineer

Jul 2022 – Aug 2025

Led the design, development, and optimization of scalable enterprise-grade backend services, distributed systems, and APIs using Python (Django, FastAPI), Go, Java, and Node.js in cloud-native environments. Delivered high-performance, full-stack

applications with React, Next.js, and TypeScript, integrating seamlessly with healthcare, finance, and AI-driven platforms. Built and deployed systems across AWS, GCP, and Azure, ensuring reliability, observability, scalability, and compliance while following security best practices.

Backend Engineering (Python, Django, FastAPI, Go, Java, Node.js, TypeScript)

- Designed and maintained backend services supporting scalable APIs, authentication, and enterprise-grade workflows.
- Implemented RESTful, GraphQL, and gRPC APIs with optimized queries in PostgreSQL, MySQL, and NoSQL databases (MongoDB, DynamoDB).
- Built ETL pipelines to process structured/unstructured data, ensuring reliable ingestion, transformation, and validation of complex biomedical and financial datasets.
- Applied security best practices (authentication, encryption, OWASP, audits), IAM policies, secrets management, and threat modeling to protect sensitive healthcare and financial data.
- Emphasized maintainable, well-structured, and well-documented code practices to ensure long-term reliability and team collaboration.

APIs, Scalability & Reliability

- Enhanced scalability with database tuning, caching, and horizontal scaling; applied event-driven patterns (Kafka, Pub/Sub, RabbitMQ, AWS SQS).
- Designed and optimized storage for time-series data using Prometheus and database extensions (e.g., TimescaleDB) for querying and analysis of telemetry and biometric data.
- Conducted load testing, chaos testing, and resilience engineering practices to validate fault tolerance under high-throughput workloads.
- Developed unit and integration test suites using pytest, unittest, and Jest to achieve high coverage and confidence in production deployments.
- Built ephemeral environments and infrastructure integration tests to strengthen CI/CD reliability.

Cloud Services & Infrastructure

- Deployed and maintained systems across AWS (EC2, ECS, Fargate, API Gateway, RDS, S3, Lambda, CloudWatch, IAM, SecurityHub), GCP, and Azure.
- Designed infrastructure-as-code workflows with Terraform, AWS CDK, and CloudFormation.
- Built CI/CD pipelines with GitHub Actions, GitLab CI, Jenkins, CircleCI, and ArgoCD.
- Containerized applications with Docker and orchestrated deployments via Kubernetes and ECS/Fargate.

Monitoring & Observability

- Implemented metrics, logging, and tracing with Prometheus, Grafana, ELK, Datadog, New Relic, OpenTelemetry, and Sentry.
- Built alerting and incident response systems for high availability and quick recovery.

AI & ML-Powered Product Development

- Built AI-powered recommenders, classifiers, and product-facing features using embeddings, semantic search, and LLM-based UX patterns.
- Prototyped and shipped AI/ML features end-to-end—from model design and integration to production deployment and iteration based on user feedback.
- Deployed models using MLFlow and AWS SageMaker; integrated Weights & Biases for experiment tracking and monitoring.
- Partnered with AI platform teams to integrate ML pipelines into production infrastructure, ensuring scalability, observability, and reliability of deployed systems.

Frontend & Full-Stack Contribution (React, Next.js, TypeScript, Angular, Tailwind CSS)

- Built high-performance, responsive applications leveraging server-side rendering (Next.js) and modular component design (React/Angular) to optimize load times, accessibility, and SEO.
- Implemented reusable design systems and UI components with Tailwind CSS and TypeScript, ensuring consistency, maintainability, and rapid feature development across projects.

ITERON

Staff Backend Engineer

Chennai, India

Oct 2019 – Dec 2021

Delivered secure, scalable, and AI-augmented full-stack SaaS platforms for financial and compliance-intensive domains, with a strong emphasis on cloud-native engineering, real-time collaboration, recommendation systems, and developer experience. Built and maintained microservices and backend tooling using Java (Spring Boot), Spring Cloud, Spring DataFlow, Node.js, TypeScript, GraphQL, React, Golang, and Python, deployed on Kubernetes (EKS, AKS, ECS) and integrated with AWS Lambda, S3, Kafka, MongoDB, Cassandra, and PostgreSQL.

Content Creation & Collaboration Systems

- Led development of microservices and real-time APIs supporting multi-user content creation, editor workflows, and presence tracking, using GraphQL, WebSockets, and MongoDB change streams.
- Built backend features to power real-time collaborative editing, draft autosaving, and sync conflict resolution, ensuring data integrity and seamless UX.

AI-Powered Features & Developer Tooling

- Integrated LangChain, LangFuse, OpenRouter, and Mastra to enable AI-assisted content recommendations, automated code/test generation, and natural language prompts for search and summarization.
- Built custom VSCode extensions to streamline accessibility checks, test coverage surfacing, and code scaffolding directly within developer IDEs.

Recommendation Engines & Pipelines

- Developed offline pipelines for personalized content and scheduling suggestions using Apache Airflow, MongoDB, and S3, with rule-based and collaborative filtering approaches.
- Integrated pipelines into GraphQL APIs powering dashboard-level insights and user-specific content curation.
- Parsed and transformed data using XML and XSLT, supporting integration with legacy reporting and third-party analytics systems.

Platform & Monorepo Engineering

- Spearheaded migration to a centralized monorepo, consolidating microservices and shared packages (design tokens, feature flags, auth libraries) using Nx and custom CLI tools.
- Introduced codemods, lint rules, and GitHub Actions to unify code quality and onboarding across teams.

Accessibility & Frontend Systems

- Delivered frontend components with React, Next.js, TailwindCSS, and Apollo GraphQL, styled for WCAG and ARIA compliance.
- Built TypeScript-based libraries and testing utilities to automate accessibility validation via CI and manual tooling.

Microservices & Cloud Architecture

- Designed and deployed event-driven microservices using Spring Boot, Node.js, FastAPI, Kafka, RabbitMQ, SQS, and Spring Cloud Stream, with seamless OAuth2/OIDC auth.
- Implemented secure integrations with SOAP-based web services to support external data exchange and enterprise system interoperability.
- Used Terraform, Helm, and GitHub Actions to provision EKS, configure autoscaling, and automate zero-downtime deployments.

Testing & CI/CD

- Maintained >90% test coverage with robust unit, integration, and E2E test suites using JUnit, Mockito, Selenium, Jest, React Testing Library, and Playwright.
- Implemented CI pipelines with GitHub Actions, CodePipeline, and container security scanning, including compliance with CIS Kubernetes Benchmarks.

Observability & Reliability

- Integrated Prometheus, Grafana, AppDynamics, ELK, and CloudWatch for full-stack observability, tracing via OpenTelemetry, and SLO-based alerting.
- Contributed to on-call rotation, led postmortems, and implemented chaos testing and retry logic for critical workflows.

Mentorship & Cross-Team Collaboration

- Mentored engineers across multiple teams on secure coding, accessible UIs, test strategy, and microservice design.
- Partnered with Product and Design to prioritize features aligned with user goals and technical feasibility.

HERZUM

Chicago, IL

Lead Full Stack Engineer

May 2018 – Dec 2018

Engineered high-resilience, observability-first platforms for large-scale SaaS and financial services, focusing on Kubernetes, Terraform, Java (Spring Boot), and full-stack delivery using TypeScript, Node.js (microservices), and React. Deployed secure, scalable systems on AWS (Lambda, S3, ECS), enhanced developer experience through CLI and IDE tooling, and upheld digital accessibility by applying WCAG, ARIA, and secure web programming best practices.

- Developed and maintained TypeScript libraries and Node.js APIs that enabled internal teams to scan and resolve accessibility issues in their codebases.
- Built and extended developer tooling including CLI tools, VSCode extensions, and integrations with testing frameworks to streamline accessibility and code quality enforcement.
- Led microservice deployments using Java (Spring Boot) and Node.js (TypeScript microservices), containerized via Docker, deployed with Helm on Kubernetes (EKS/AKS), and monitored using Prometheus and Grafana.
- Designed and maintained Terraform modules for managing compute, networking, and Kubernetes resources, enabling repeatable, secure infrastructure as code deployments.
- Created unit tests, integration tests, and CI/CD workflows using GitHub Actions, embedding security scanning, accessibility linting, and test automation across services and tools.
- Designed and optimized relational schemas for PostgreSQL to support transactional workflows and batch analytics pipelines.
- Tuned PostgreSQL and Kafka consumers for high-volume transaction processing, including rebalancing and I/O optimizations for sustained throughput under load.
- Integrated with messaging systems such as Kafka and SQS to support asynchronous processing, retry logic, and event-driven architectures.
- Delivered and maintained RESTful and GraphQL (Apollo) APIs supporting both internal systems and external clients, emphasizing performance, accessibility, and secure data access.
- Collaborated with product managers to define and prioritize product features and accessibility improvements, contributing to the roadmap for platform tooling.
- Authored and reviewed technical proposals to enhance platform architecture and developer tooling infrastructure.
- Applied best practices in secure web development, secrets management, and static code analysis, aligning with DevSecOps principles.
- Developed and managed Python-based ETL workflows using Apache Airflow and data transformation with Pandas, ensuring reliable, scalable data pipelines.
- Leveraged AI tools such as LangChain, LangFuse, and OpenRouter to prototype and integrate AI-powered workflows into product features.
- Utilized DynamoDB and Elasticsearch alongside relational databases to support distributed data needs.
- Implemented frontend UI architecture with React, Tailwind CSS, and SingleSPA for modular, scalable user interfaces.

ACCENTURE

Naperville, IL

Lead Full Stack Engineer

Jul 2017 – Apr 2018

Designed and implemented secure, scalable API gateways and microservices architectures using Java (Spring Boot), Go, Node.js, and TypeScript, deployed on Kubernetes (GKE/AKS) with Istio service mesh for zero-trust security and fine-grained policy enforcement. Developed accessible frontend components using ReactJS, integrating accessibility standards such as WCAG and ARIA to ensure inclusive user experiences. Championed developer productivity by building and maintaining tooling libraries, VSCode extensions, CLI tools, and automated testing integrations to help teams detect and fix accessibility issues early.

- Led microservices deployments in Kubernetes with horizontal pod autoscaling, resource optimization, and container security best practices, including image scanning and least privilege roles.
- Developed and maintained TypeScript products and Node.js libraries that support accessibility checks and developer tooling integrations for axe Monitor and other accessibility platforms.
- Integrated OpenTelemetry for distributed tracing, enabling deep insights into authentication flows and master data synchronization workflows.

- Hardened CI/CD pipelines by embedding automated security gates and vulnerability scans using GitHub Actions and CircleCI, ensuring robust software delivery.
- Supported messaging-driven architectures using RabbitMQ and MongoDB for real-time master data syncs and event-driven workflows.
- Developed a secure AWS Lambda proxy layer in Python for token validation and traffic shaping, integrated with Amazon API Gateway, ensuring scalable and secure cloud APIs.
- Extended Java-based authorization services leveraging AWS IAM roles, STS tokens, and Secrets Manager to implement fine-grained, environment-specific access control.
- Authored and reviewed technical proposals to improve product architecture, infrastructure, and accessibility tooling workflows.
- Created Python toolkits for automated health diagnostics and incident remediation, deployed securely via AWS Systems Manager.
- Participated in thorough pull request reviews, including security, accessibility, and code quality assessments, fostering knowledge sharing and team growth.
- Collaborated closely with global Agile teams, Product Managers, and Customer Success to prioritize feature development, bug fixes, and user support.

WAVICLE DATA SOLUTIONS

Oakbrook, IL

Data Engineer

Apr 2017 – Jun 2017

Developed and supported full-stack applications using Java, Python, and Scala in Agile teams. Built distributed microservices and real-time data pipelines with Kafka and Spark. Delivered robust cloud solutions on AWS and Snowflake.

- Designed and deployed real-time data pipelines using Kafka and Spark, reducing data processing latency by 50% and enabling near-instant analytics for business users.
- Delivered scalable full-stack applications with Java, Python, and Scala, improving application performance by 35% and reducing deployment time through CI/CD automation.
- Architected and implemented cloud-based data workflows on AWS and Snowflake, resulting in a 40% increase in data availability and reporting accuracy.

JPMORGAN CHASE

Chicago, IL

Sr. Application Developer

Jan 2017 – Mar 2017

Played a key role in the accelerated development and deployment of a mission-critical financial analytics application in a dynamic enterprise setting. Focused on optimizing application performance, enhancing system integration, and improving database efficiency.

- Enhanced application response times by 40% through performance tuning and advanced query optimization using Postgres and Elasticsearch.
- Integrated 5 external and internal APIs, leveraging gRPC and GraphQL, to increase data accessibility by 50% through close collaboration with cross-functional teams.
- Automated backend workflows using AWS Lambda, Step Functions, EventBridge, and managed scalable data storage with AWS DynamoDB.
- Utilized Kafka for robust, event-driven communication between services, improving system reliability and decoupling components.
- Employed the AWS Cloud Development Kit (CDK) to streamline infrastructure as code, accelerating deployment cycles and enhancing environment consistency.
- Refactored core data processing modules in Java to integrate with Amazon SQS and SNS, enabling asynchronous transaction tracking and alerting.
- Developed ETL pipelines in Python using AWS Glue, S3, and Boto3, enabling automated ingestion of financial datasets and compliance reporting.
- Implemented role-based access control for microservices using AWS Cognito, integrating with Java-based authentication services for secure, token-based workflows.

INFOGIX INC

Naperville, IL

Sr. Application Developer

Oct 2014 – Dec 2016

Led full lifecycle development of scalable enterprise SaaS solutions, emphasizing client usability, performance, and operational efficiency. Partnered with product management and engineering teams to deliver impactful features while spearheading DevOps efforts to optimize deployment processes. Played a pivotal role in client onboarding and product training to enhance adoption and customer satisfaction. Led MDM schema design and reconciliation workflows to reduce duplication and enforce referential integrity. Built custom data validation pipelines using AWS Step Functions and SQS for async batch processing. Developed RESTful APIs to expose consistent views of master data across client-facing services. Tuned PostgreSQL with partitioning, query plan analysis, and indexing strategies to support 100M+ records. Established MDM governance standards, including lineage tracking and update history. Integrated automated data remediation and alerting using Grafana and Prometheus.

- Designed and implemented CI/CD pipelines using cloud-native tools, achieving a 60% reduction in deployment times and enabling more frequent, reliable releases.
- Developed a comprehensive client dashboard feature utilizing Java, J2EE, and APIs built with gRPC and GraphQL, which reduced client support tickets by 40%.
- Automated backend workflows and event-driven processes with AWS Lambda, Step Functions, EventBridge, and maintained scalable storage with AWS DynamoDB and Postgres.
- Engineered real-time data pipelines leveraging Kafka for asynchronous communication and used Elasticsearch to improve search capabilities and query performance.
- Utilized the AWS Cloud Development Kit (CDK) to automate infrastructure provisioning, accelerating deployment cycles and improving environment consistency.
- Led MDM schema design and reconciliation workflows to reduce duplication and enforce referential integrity.
- Built custom data validation pipelines using AWS Step Functions and SQS for async batch processing.
- Developed RESTful APIs to expose consistent views of master data across client-facing services.
- Tuned PostgreSQL with partitioning, query plan analysis, and indexing strategies to support 100M+ records.
- Established MDM governance standards, including lineage tracking and update history.
- Integrated automated data remediation and alerting using Grafana and Prometheus.

MCDONALD'S

Oakbrook, IL

Platform Engineer – Event-Driven Authorization & Data Services

May 2012 – Sep 2014

Built distributed authorization and identity event pipelines for mission-critical logistics applications. Designed secure, reactive services in Go, Kafka, and PostgreSQL to support high-frequency access events.

- Modeled RBAC and ABAC policies in a dynamic evaluation engine for complex organization hierarchies.
- Built event-driven data ingestion for MDM changes using Kafka Streams and CDC.
- Improved PostgreSQL write performance using batching and index optimizations during peak loads.
- Integrated Prometheus and Loki to monitor and log API token usage, failed authentications, and permission errors.
- Developed real-time dashboards to track role assignment changes and permission propagation delays.
- Collaborated with SRE teams to test failover strategies and ensure platform resilience across availability zones.

VESTAS

Chennai, India

Assistant Lead Engineer

Sep 2011 – May 2012

Contributed to the development and optimization of wind turbine systems through advanced algorithm design and simulation modeling. Focused on enhancing operational efficiency and reducing research and development overhead. Performed large-scale performance analyses across wind farms and supported continuous improvement efforts. Provided technical guidance to junior engineers to improve team capability and delivery output.

- Designed optimization algorithms that improved wind turbine energy efficiency by 18%.
- Reduced R&D costs by 20% by developing innovative simulation models for performance testing.
- Conducted performance analysis for over 30 wind farms, identifying multiple efficiency improvement opportunities.
- Mentored 5 junior engineers, increasing team productivity by 10%.

CATERPILLAR

Peoria, IL

Assistant Lead Engineer

Dec 2009 – Aug 2011

Led major enterprise IT modernization efforts focused on legacy system migration, infrastructure optimization, and data engineering. Directed cross-functional analyst teams to ensure on-time delivery of high-quality technology solutions aligned with business goals. Played a key role in implementing scalable ETL pipelines and system upgrades to improve performance, reduce costs, and support operational continuity. Led a team of 10 analysts, ensuring the timely delivery of high-quality, scalable solutions.

- Spearheaded the migration of 50+ legacy applications to a modern ERP system, resulting in \$1M annual cost savings.
- Improved system uptime to 99.95% through targeted infrastructure upgrades and performance tuning.
- Reduced data processing times by 35% by implementing a highly efficient ETL pipeline.

CAPITAL GROUPS

Irvine, CA

Programmer Analyst

Mar 2006 – Nov 2009

Focused on enhancing investment management systems and streamlining operational workflows through custom application development and automation. Worked closely with portfolio managers and operations teams to deliver high-impact tools that improved efficiency and decision-making. Proactively addressed system issues to ensure minimal disruption and high client satisfaction.

- Developed portfolio management tools that improved investment decision-making efficiency by 30%.
- Automated reporting processes, saving over 5,000 hours annually for the operations team.
- Delivered a major application upgrade project one month ahead of schedule, avoiding \$250K in contractual penalties.
- Achieved 99.8% client satisfaction by quickly identifying and resolving critical system issues.

REUTERS

St. Paul, MN

Software Programmer

Jan 2006 – Feb 2006

Contributed to a fast-paced software development initiative focused on enhancing news data aggregation and improving application performance. Collaborated with cross-functional teams to rapidly address stability issues and deliver an intuitive user experience within a tight project timeline.

- Developed a news aggregation tool that reduced data retrieval time by 50%.
- Resolved critical bugs within 3 weeks, improving overall application stability by 20%.
- Collaborated with UI/UX teams to deliver a user-friendly interface, increasing usability by 15%.

MOBILE APPS DEVELOPMENT (IOS AND ANDROID)

Responsible for every aspect of design and development, from conceptualization to implementation. Developed six apps using SwiftUI and developed two apps (iOS and Android versions) using Flutter. Leveraged Google Cloud Firebase API: Used Firebase Firestore and Firebase Storage for Nerding app, GuessedRight, Poll Hippo, and Socically. Used Firebase Auth for Nerding app (App Store and Play Store), Poll Hippo, and Socically.

- Developed 6 iOS apps (QuizzlyMath5-WholeNumbers, Sequenced, GuessedRight, nerding app, Poll Hippo, Socically) currently available in the AppStore.
- Developed 1 Android app (nerding app) currently available in the PlayStore.